MAJOR PROJECT ASSIGNMENT

Module Design, Flowchart, UML Diagrams & Data Flow Diagram

Multimodal Based Authentication for High Security Applications
Risk-Aware Cross-Modal Consistency Fusion Using Face and Voice Biometrics

Student Name	_____
USN / Register No.	_____
Class / Section	_____
Guide / Faculty	_____
Submission Date	_____

This document presents a proposed design based on the project overview supplied by the team. The diagrams explain the system workflow, software interactions and movement of biometric data. They are design representations, not a claim that every module is already implemented.

1. Introduction and Design Overview

The project combines face and voice biometric evidence to authenticate a user. A conventional fixed-weight fusion method may give the same importance to both modalities even when one input is poor or suspicious. The proposed design considers modality quality, liveness evidence and face–voice consistency before determining risk and adapting the fusion decision.

The project-specific contribution is the implementation and evaluation of a risk-aware cross-modal consistency mechanism. The project does not claim that face–voice authentication, liveness or adaptive fusion are newly invented concepts.

Main objectives

• Capture and preprocess face and voice inputs.
• Extract face and voice representations and generate independent matching scores.
• Estimate quality and liveness/reliability for each modality.
• Measure consistency between face and voice evidence.
• Calculate a risk indicator and adapt the contribution of each modality.
• Compare the proposed method with conventional fixed-weight fusion.

Technology mapped to the design

Layer / module	Technology in project overview	Responsibility
Frontend	React.js + Vite	Registration/login, capture/upload and result display.
Backend/API	FastAPI	Connect frontend requests to biometric and authentication modules.
Face processing	OpenCV; DeepFace / FaceNet-style embeddings	Face detection, preprocessing, representations and matching.
Voice processing	Librosa + soundfile; MFCC and spectral features	Audio loading, normalization, feature extraction and matching.
ML / custom logic	PyTorch; custom Python modules	Consistency, risk calculation and adaptive fusion.
Quality / liveness	OpenCV + Librosa measures; modality-specific checks	Estimate usability and identify suspicious/non-live evidence.
Storage / versioning	MongoDB; Git + GitHub	Store protected references/logs and track code changes.

Exact model selection, liveness methods, risk coefficients and thresholds are to be finalized and experimentally validated.

2. Detailed Module Design

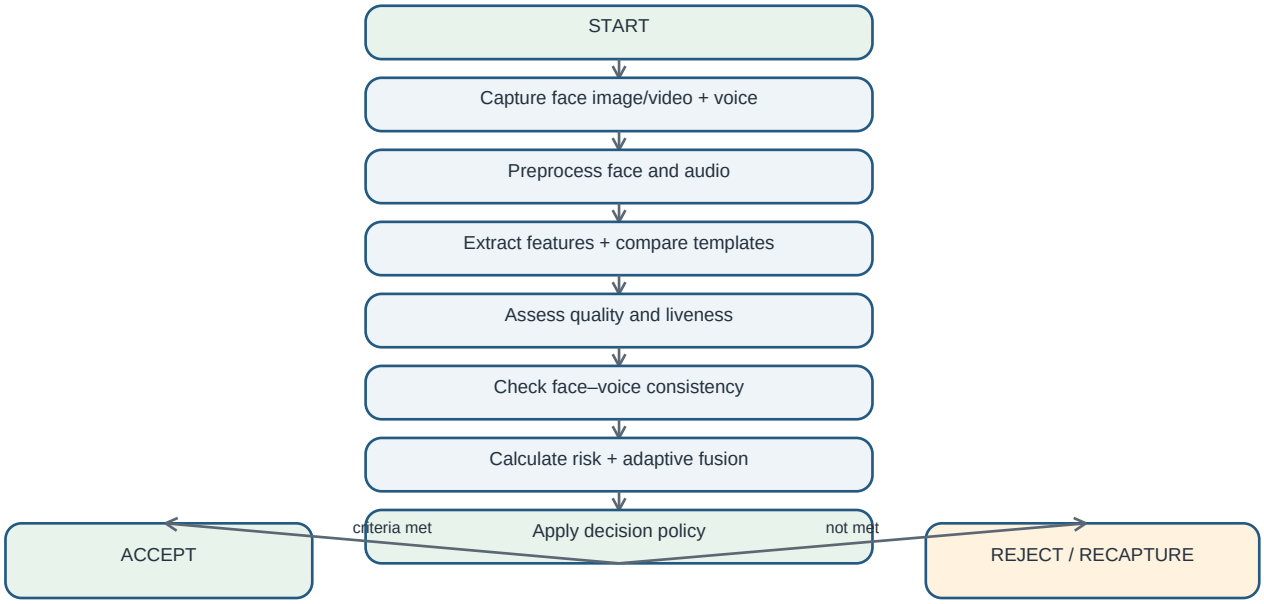
The system is divided into modules so that each part has a clear responsibility. Modules exchange features, scores and status information through defined interfaces.

Module	Input	Main processing	Output
1. User interface	Registration details; face image/video; voice sample	Provide screens, validate required fields, send requests and display results.	Backend request; result shown.
2. API / backend	Frontend request and biometric data	Validate payloads and coordinate module execution through APIs.	Processing request; response.
3. Face processing	Face image/video frames	Detect face, preprocess/resize frames and generate embedding using selected model.	Face features and matching score.
4. Voice processing	Voice/audio sample	Load and normalize audio; extract MFCC/spectral features or selected embedding; compare template.	Voice features and matching score.
5. Quality assessment	Face frames and audio signal	Estimate whether each sample is sufficiently usable.	Face and voice quality values.
6. Liveness assessment	Face/video and voice evidence	Apply modality-specific anti-spoofing/liveness checks.	Liveness evidence/status.

Module	Input	Main processing	Output
7. Cross-modal consistency	Paired face and voice evidence	Estimate whether both modalities are consistent for the same attempt.	Consistency score.
8. Risk engine	Quality, liveness and consistency values	Calculate modality reliability and overall risk using configured coefficients.	Risk and reliability values.
9. Adaptive fusion	Face/voice scores, reliability, risk and consistency	Assign modality weights according to reliability and combine scores; apply risk/consistency gate.	Fused score and risk state.
10. Authentication decision	Fused score, threshold and risk state	Apply policy; accept, reject or request fresh capture.	ACCEPT / REJECT / recapture.
11. Database / logging	Registration references, metadata and result	Store required protected references and relevant authentication logs.	Reference data and log records.

3. Flowchart – Authentication Workflow

The flowchart represents the order of operations in one authentication attempt. It starts with capturing both biometric inputs and proceeds through preprocessing, feature extraction, quality/liveness checks, consistency analysis, risk calculation and adaptive fusion.



Explanation

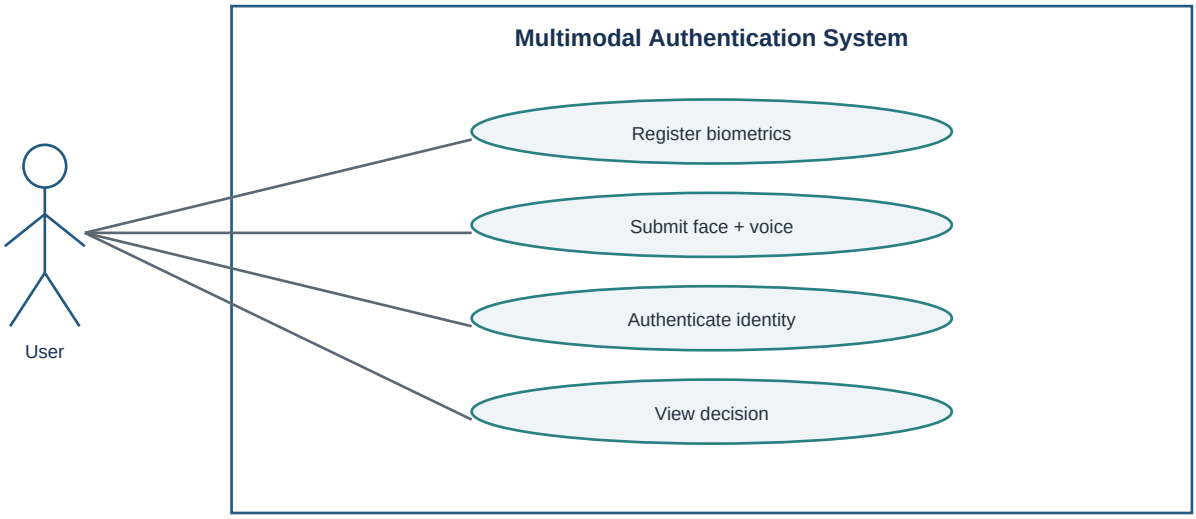
1. Start and capture: the user supplies a face image/video and voice sample.
2. Preprocess: prepare the face and audio inputs for analysis.
3. Feature extraction and matching: compare face and voice representations with registered templates to obtain independent scores.
4. Quality and liveness: estimate usability and check available anti-spoofing indicators.
5. Cross-modal consistency: check whether face and voice evidence agree for the same attempt.
6. Risk and fusion: reliability and risk values influence the face and voice weights.
7. Decision: the configured policy returns ACCEPT or REJECT; a fresh capture may be requested if evidence is insufficient.

The exact threshold values and branch conditions are implementation parameters and must be validated with the selected dataset and test cases.

4. UML Diagrams

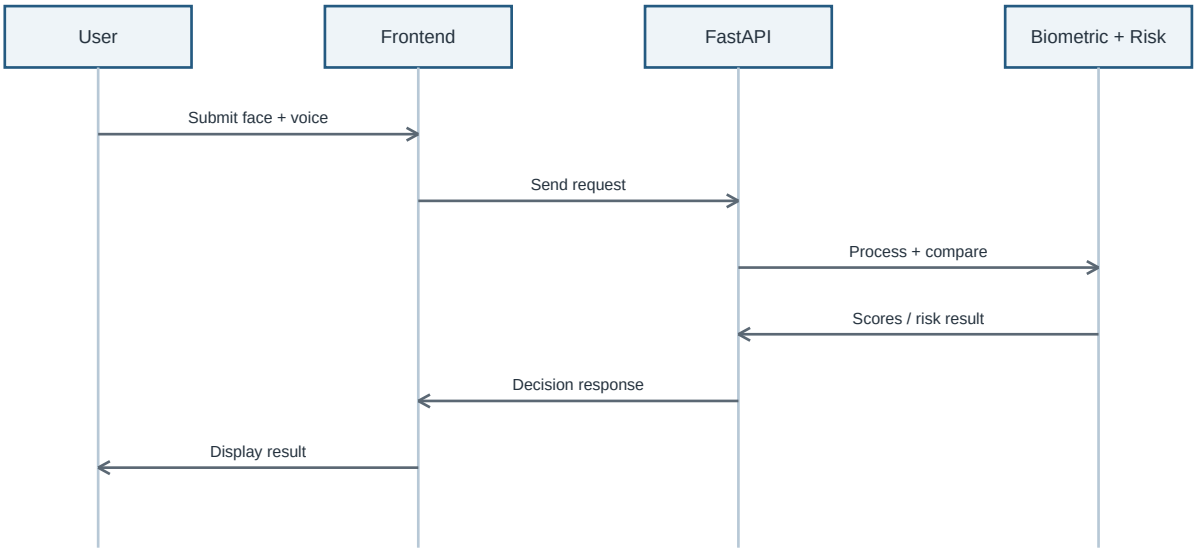
UML (Unified Modeling Language) diagrams describe the system from different viewpoints. The following diagrams are relevant to this project: use-case, sequence and component diagrams.

4.1 Use-Case Diagram



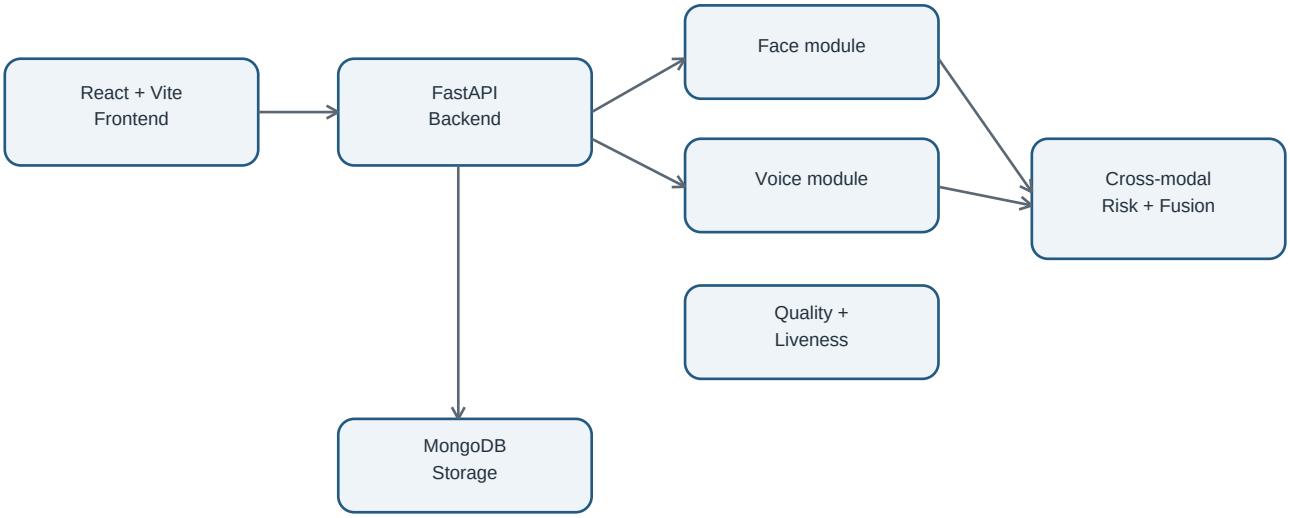
Explanation: The user is the main actor. The user can register biometric references, submit face and voice inputs, request authentication and view the result. An authorized administrator or reviewer may inspect permitted authentication logs. This diagram shows user-visible functions rather than the internal algorithm.

4.2 Sequence Diagram



Explanation: The user submits biometric inputs through the frontend. The frontend sends the request to FastAPI. Backend processing invokes biometric and risk modules. The modules return matching scores, quality/liveness evidence and risk/fusion results. The backend returns the decision to the frontend, which displays it to the user.

4.3 Component Diagram

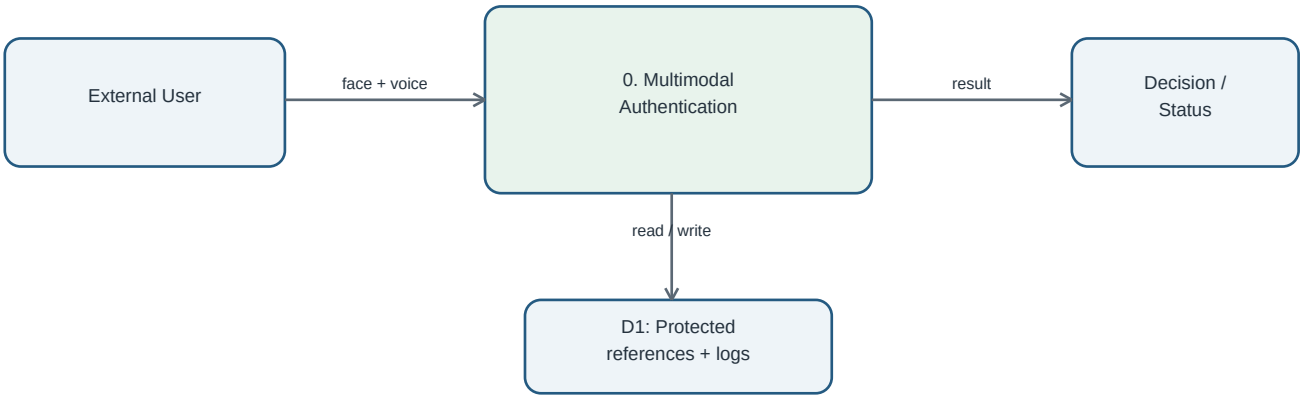


Explanation: React/Vite provides the interface; FastAPI connects it to processing modules. Face and voice modules produce biometric evidence. Quality/liveness and cross-modal analysis feed the risk and fusion logic. MongoDB is planned for protected references, metadata and authentication logs. Internal API boundaries may be refined during implementation.

5. Data Flow Diagram (DFD)

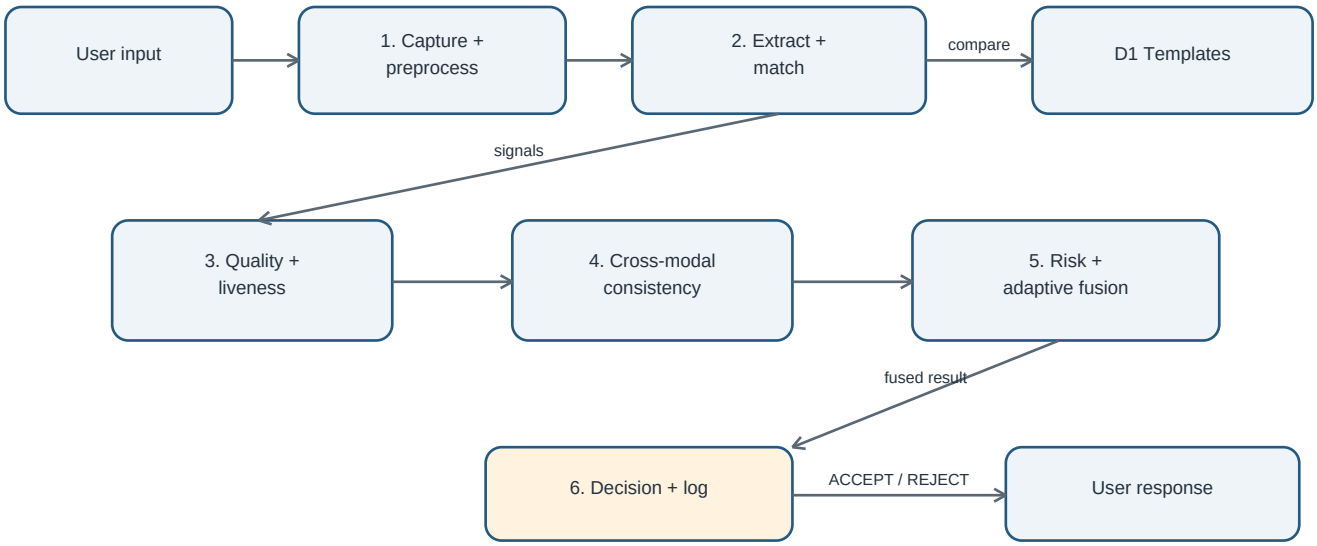
A Data Flow Diagram represents how information enters the system, moves between processes, interacts with data stores and leaves the system. The diagrams below show a context-level view (Level 0) and a detailed view (Level 1).

5.1 DFD Level 0 – Context Diagram



Explanation: The user provides face and voice inputs to the overall authentication process. The system uses stored biometric references where required and returns an authentication status. The data store represents protected references and logs.

5.2 DFD Level 1 – Detailed Data Movement



Explanation

1. Capture/preprocessing: incoming face/video and audio are prepared for analysis.
2. Feature extraction/matching: representations are generated and compared with registered templates.
3. Quality/liveness: usability values and liveness evidence are produced.
4. Cross-modal consistency: face and voice evidence are evaluated together.
5. Risk/adaptive fusion: matching scores are combined using reliability-aware weights and risk/consistency information.
6. Decision/log: the policy produces the result, relevant records may be stored and the result is returned to the user.

6. Key Data Items and Interfaces

These are logical data items based on the proposed architecture. Final field names and formats will be decided during implementation.

Data item	Produced by	Used by	Meaning
Face input / frames	Frontend / capture	Face, quality, liveness	Image or video for the current attempt.
Voice input / audio	Frontend / capture	Voice, quality, liveness	Audio sample for the current attempt.
Face embedding / score	Face module	Risk / fusion	Face representation and matching result.
Voice features / score	Voice module	Risk / fusion	Voice representation/features and matching result.
Quality values	Quality module	Risk engine	Estimated usability of both samples.
Liveness evidence	Liveness module	Risk / decision	Indicators of live or suspicious presentation.
Consistency score	Cross-modal module	Risk / fusion gate	Agreement between face and voice evidence.
Risk / reliability	Risk engine	Adaptive fusion / decision	Values controlling modality contribution.
Fused score	Adaptive fusion	Decision module	Combined score using adaptive weights.
Decision / log	Authentication module	Frontend / database	Final status and relevant authentication record.

7. Risk-Aware Fusion Logic

Let S_face and S_voice be normalized matching scores; R_face and R_voice be modality reliability values; and C_cross be normalized cross-modal consistency. The project overview gives the following example reliability expression:

$$R_{overall} = \alpha R_{face} + \beta R_{voice} + \gamma C_{cross}$$

Example adaptive weights:

$$w_{face} = R_{face} / (R_{face} + R_{voice})$$
$$w_{voice} = R_{voice} / (R_{face} + R_{voice})$$

Fused score:

$$S_{fused} = w_{face} \times S_{face} + w_{voice} \times S_{voice}$$

Cross-modal consistency and risk may also act as a gate. If consistency is too low or risk exceeds a configured threshold, the system may reject the attempt or request a fresh capture. Coefficients, normalization, zero-reliability handling and thresholds must be defined and validated experimentally; these equations are a starting point, not final measured settings.

8. Assumptions and Design Notes

- The user has registered face and voice references before normal authentication.
- Face and voice matching are first evaluated independently.
- Quality, liveness and consistency signals are inputs to the risk-aware decision layer.
- The database is intended to store protected references, required metadata and authentication logs.
- Evaluation should include good-quality, low-quality, inconsistent, failed-liveness and repeated-attempt conditions.
- Fixed-weight fusion should be compared with risk-aware fusion under the same test conditions.

9. Conclusion

The module design separates acquisition, biometric processing, quality/liveness assessment, cross-modal consistency, risk calculation, adaptive fusion and authentication. The flowchart explains the processing order, UML diagrams describe user interactions and software structure, and the DFD explains how biometric information moves through the system. Together,

these diagrams provide a design-level blueprint for implementation and testing.

Source basis: the team's project overview, "Multimodal Based Authentication for High Security Applications – Risk-Aware Cross-Modal Consistency Fusion." The diagrams are proposed design representations of that overview; they do not claim every module is already implemented.